PakEmis Sector Technical Note TN04

Three Categories Partitioning One Waste Stream: Solid-Waste Disposal, Open Burning and Domestic Wastewater Emissions for Pakistan

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Abstract

Pakistan's waste sector has no current national greenhouse-gas estimate built from Pakistani activity data, and the international products that fill the gap disagree with each other by amounts larger than the mitigation decisions they inform. We report the sector for FY2023–24 at 37.45 Mt CO_2e (AR5, 36.37 Mt on AR6) across three IPCC categories: domestic wastewater (4D1) at 24.46 Mt, solid-waste disposal (4A) at 9.17 Mt and open burning (4C2) at 3.83 Mt. The sector is 8.4 % of a ten-sector national inventory and, after domestic aviation (0.1 % of the total at ±58.8 %), carries its second-widest uncertainty band: ±51.08 % by Approach-1 quadrature, and 25.7 Mt to 61.4 Mt (2.5th–97.5th percentile) under the parameter-rerun Monte Carlo, whose deliberate asymmetry prices the effluent N_2O factor's 150-fold published range that a symmetric band cannot carry.

Two features distinguish the estimate from a conventional compilation. First, solid-waste disposal and open burning are computed from one mass-balanced national waste-generation figure rather than from separate literature: 31.56 Mt generated, 50.3 % collected to disposal and the remainder uncollected, of which 60 % burns and the unburned 6.27 Mt is carried as an explicit unresolved fate with no emissions pathway applied. Disposed plus uncollected equals generated, asserted with exact equality rather than reconciled afterwards, so an argument about the collection rate moves emissions between two categories without creating or destroying waste. Inventories that estimate landfilling and open burning independently have no such constraint, and nothing then forces their sum to be a quantity of waste the country actually produces.

Second, the largest category rests on an assumption the guidelines leave genuinely open for Pakistani conditions. IPCC gives a methane correction factor of 0.5 for stagnant sewers and 0.1 for discharge to a flowing river; Pakistani urban drainage has properties of both. At 0.5 domestic wastewater is 24.46 Mt and the sector sits at 141 % of EDGAR's year-matched Pakistani waste total; at 0.1 it is 15.39 Mt and the sector lands close to EDGAR. The physical description of the drains supports the higher value and the only independent comparator supports the lower. We report both, adopt 0.5 as central because the physical description is the stronger evidence, and identify the measurement that would settle it: dissolved-methane sampling on urban drains across a seasonal cycle. Given that this single category carries 13.0 % of national inventory variance under Approach 1 while contributing 5.5 % of national emissions, that is among the highest-return measurements available to Pakistan's inventory.

Keywords: municipal solid waste, first order decay, open burning, domestic wastewater, methane correction factor, exhaustive partition, IPCC 2006 Guidelines, Monte-Carlo uncertainty, Pakistan, MRV

PakEmis Sector Technical Note TN04. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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Release provenance. This report documents one release of PakEmis and is generated against it. Where a sector report and the release's generated product catalogue (`outputs/pakemis_product_catalog.csv`) disagree, the catalogue is authoritative: sector reports are revised on their own cadence and may lag the products.

Release version	`1.0.0rc67`
Commit	`1876f398f85c`
Reporting period	FY2023-24 (2023-07-01 to 2024-06-30)
Baseline configuration	S (survey basis)
National total	444.96 Mt CO_2e (AR5) under S, the reference; 450.20 Mt under L
Release manifest	`13462fb4c2d1ea62` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

Waste is the category of a national greenhouse-gas inventory where the arithmetic is easiest and the activity data hardest. The emission factors are few and well characterised; what is difficult is knowing how much waste a country generates, what fraction of it reaches a disposal site, what happens to the fraction that does not, and how the sewage of a large population is actually handled. None of those quantities is metered anywhere in Pakistan.

That difficulty has a specific consequence for how the sector is usually compiled. Because the activity data must be modelled rather than read, each category tends to be estimated on its own terms: landfill methane from a tonnage assumption, open burning from a per-capita burning rate in the literature, wastewater from a population and a default organic load. The three estimates are then reported together. Nothing in that procedure requires the waste sent to landfill and the waste openly burned to sum to a quantity of waste the country plausibly produces, and in general they do not.

1.1. What this paper does

We report Pakistan's waste sector for FY2023–24 from Pakistani activity data, with the three categories constructed so that the constraint the conventional procedure lacks is enforced by construction. A single national waste-generation figure is partitioned: what is collected goes to the first-order-decay model for solid-waste disposal, what is not collected is what can burn, and the two sum to the generated quantity exactly. The partition is asserted in code with equality rather than checked afterwards with a tolerance, because a tolerance permits precisely the drift the constraint exists to remove.

The result is a sector total of 37.45 Mt CO_2e, of which domestic wastewater is two-thirds. It is the second most uncertain sector in the national inventory that contains it, and this paper is as much about where that uncertainty lives and what would reduce it as about the central estimate.

1.2. Why Pakistan, and why this sector now

Pakistan is the fifth most populous country in the world, urbanising rapidly, and its most recent full national inventory submission reports for FY2017–18 (Ministry of Climate Change, Government of Pakistan, 2022). In the interim the international gridded products—EDGAR (Crippa et al., 2025, 2020) and Climate TRACE (Climate TRACE, 2026)—supply the numbers that inform policy. They are global models applying globally consistent assumptions to nationally reported activity, which is their strength for cross-country comparison and their limitation for any one country: where a national compiler can consult a sanitation survey or a district population panel, a global product must choose a default.

The waste sector is where that difference bites hardest, for two reasons. Its activity data are the least standardised internationally, so the defaults do the most work. And it is dominated by methane, whose short atmospheric lifetime makes it the gas where near-term mitigation is most effective and therefore where the estimate most directly changes what a government should do.

1.3. Contributions

1. **A current national waste-sector inventory for Pakistan** covering solid-waste disposal, open burning and domestic wastewater, from a district population panel and national sanitation statistics rather than from a global model's Pakistani slice.
2. **One waste stream, mass-balanced across its fates**: collected and uncollected shares sum to generation exactly, two fates carry emissions pathways (4A, 4C2) and the unburned remainder ships as an explicit unresolved row rather than a silent omission. The closed balance removes a class of inconsistency that independent estimation cannot detect (Section 4).
3. **Both readings of the sewer methane correction factor**, with the evidence for each stated and neither resolved by fiat, together with the measurement that would settle it (Section 9).
4. **A spin-up sensitivity for the decay model**, quantifying how much of today's landfill methane comes from waste deposited before the activity panel begins—a question most first-order-decay implementations answer implicitly and do not report.
5. **An air-pollutant layer on the same activity basis** as the greenhouse gases, so that open burning's particulate can be compared with other Pakistani sources without the usual mismatched-activity problem.

Method that this sector shares with the other nine of the national inventory—the module contract, the uncertainty treatment, the notation keys, the accounting closures—is documented in the national manuscript (Ahmad, 2026) and not repeated here.

2. Related Work

2.1. Global products and what they can and cannot resolve

EDGAR (Crippa et al., 2025, 2020) reports a Pakistani waste total covering disposal, burning and wastewater together, and is the closest external comparator to the scope reported here. Climate TRACE (Climate TRACE, 2026) takes a different approach, inferring emissions from asset-level and remote-sensing evidence. Both are global products: they apply a consistent methodology across every country, which is what makes them comparable and also what limits them nationally. A global waste model must assume a collection rate and a sewer condition for Pakistan; a national compiler can consult the sanitation statistics and the district population panel that describe it.

The disagreement between products is not a defect of any of them, but it is large enough to matter. This inventory sits at 141 % of EDGAR's year-matched Pakistani waste figure, and Section 6 argues that a single assumption—the sewer methane correction factor—accounts for most of the difference.

2.2. *Waste generation and composition in South Asia*

Per-capita waste generation and its composition are the two activity inputs the sector cannot avoid, and both are surveyed rather than metered. The global compilation by Kaza et al. (2018) is the standard reference; its Pakistan generation row re-bins the one primary national survey, the Pak-EPA eight-city study (Pakistan Environmental Protection Agency and Environmental Protection and Management Consultants, 1996), whose rates are used here directly, and its collection coverage supplies the uncollected fractions. Composition matters more than it first appears: the fossil carbon fraction of municipal waste—plastics, textiles, rubber—determines how much of the CO_2 from open burning is reportable, because the biogenic remainder is a memo item under IPCC's convention rather than a national-total emission.

2.3. *First Order Decay and its implementation choices*

Landfill methane is not a function of this year's waste but of every year's waste still decaying, which is why IPCC's Volume 5 method is a first-order decay model over a deposit history rather than an emission factor applied to an annual tonnage. Three implementation choices then do most of the work and are reported inconsistently in the literature: the decay constant, the methane correction factor describing site management, and how far back the deposit series is extended before the model is considered spun up.

The third is the one this paper treats explicitly. A model started at the first year of an activity panel implicitly asserts that nothing was landfilled before it, which is never true and always biases the estimate low. The size of that bias is rarely reported. Section 7 reports it here.

2.4. *The wastewater methane correction factor*

For domestic wastewater the corresponding controversy is the methane correction factor, which translates a sanitation pathway into the fraction of its organic load that degrades anaerobically. IPCC's default table distinguishes stagnant sewers from flowing ones by an order of magnitude in that fraction, and the distinction was drawn for systems that are clearly one or the other. Open urban drains that are stagnant in the dry season and flowing after rain, and which discharge to irrigation channels, are not obviously either.

Pakistani sanitation coverage itself is well characterised: the WHO/UNICEF Joint Monitoring Programme (WHO/UNICEF Joint Monitoring Programme, 2023) reports the population shares using sewers, septic systems, latrines and open defecation, and those shares are what allocate the national organic load across pathways here. What the JMP does not report, and no Pakistani source does, is the hydraulic condition of the sewered fraction—which is the quantity the correction factor is asking about.

3. Study Area and Data

3.1. *Population, which is the activity datum*

Every category in this sector is driven by population. Waste generation is a per-capita rate applied to urban and rural residents; the organic load entering the wastewater system is a per-capita BOD applied to the same population split by sanitation pathway. There is no tonnage measurement and no metered flow anywhere in the chain, so the population panel is the activity data in a way that is unusual for an emissions inventory.

The panel is built at district level and interpolated between censuses, covering 134 districts, with the 2023 census as its reference—the census's full frame including the Islamabad Capital Territory, Gilgit-Baltistan and Azad Kashmir, a deliberately wider base than the 116 four-province districts the agriculture sector allocates to, whose activity statistics are provincial in scope. It resolves to 97 million urban and 150 million rural residents, 247 million in total. The urban/rural split is load-bearing twice over: urban residents generate more waste per head and are far more likely to have their waste collected, and the two effects run in opposite directions for emissions, since collected waste goes to a decaying landfill while uncollected waste burns.

Building the panel at district level rather than nationally is what makes the provincial allocation in Section 7 a real geography rather than a national figure split by a population share, and it is the same panel that drives the sector's air-pollutant layer.

3.2. *Waste generation and collection*

Per-capita generation is 0.453 kg per person per day urban and 0.283 kg rural, from the Pak-EPA eight-city survey (Pakistan Environmental Protection Agency and Environmental Protection and Management Consultants, 1996), the rates the Kaza et al. (2018) Pakistan row re-bins. Across the panel that is 31.56 Mt of municipal solid waste a year.

Collection is expressed as its complement, the uncollected fraction: 30 % urban and 70 % rural, which resolves to 50.3 % of national generation collected. Of the uncollected remainder, 60 % is openly burned; the rest is dumped, tipped into watercourses or otherwise disposed of without combustion and emits nothing this category reports.

These three parameters—two generation rates, two uncollected fractions and one burned fraction—are the softest inputs in the sector, and Section 10 says so. They are also the only place where a judgement moves emissions *between* categories rather than changing the total, which is a property of the partition rather than of the parameters.

3.3. *Waste composition*

Open burning needs the carbon content of what burns. Dry matter is 70 % of wet weight; total carbon is 0.2195 t per wet tonne and the fossil fraction of it is 0.0757 t per wet tonne, so roughly a third of the carbon in Pakistani municipal waste is fossil in origin. That third is what produces reportable CO_2; the biogenic remainder is a memo item, following IPCC Volume 5 Chapter 5 §5.1.

3.4. *Sanitation*

The wastewater category needs the population split across sanitation pathways, because the methane correction

factor is a property of the pathway rather than of the person. Shares come from the WHO/UNICEF Joint Monitoring Programme (WHO/UNICEF Joint Monitoring Programme, 2023), each year taking its own: sewered, septic, latrine and open defecation, each carrying its own correction factor (0.5, 0.5, 0.1 and 0 respectively).

Organic load is 40 g BOD per person per day, the regional default covering Pakistan, with an industrial-and-commercial correction of 1.25 for load discharged to the domestic system from small commercial premises. Nitrous oxide is computed from protein consumption rather than from BOD, which is why it is invariant under both of the sensitivities reported in Section 7.

3.5. What this inventory has not obtained, and what it would change

Nothing compiled here reports landfill tonnages at facility level, methane recovery at any site, industrial solid-waste quantities, or industrial wastewater organic loading. Three of those four have leads that have not been followed: the Urban Unit has mapped solid-waste services and facilities across Punjab's tehsils, PBS's district-level living-standards survey records household disposal route—municipal collection, private, public bins, roadside, waterway, open space—and PCRWR has published organic loading for named Pakistani drains. None would give complete national coverage, and each would replace a national assumption with district or facility evidence for the population it covers. The first two would replace modelled disposal with measured disposal; the second two are whole categories this inventory reports as not estimated. Section 10 takes each in turn.

4. Methods

4.1. One generation figure, mass-balanced across fates

The sector's structure follows from a single decision: municipal solid waste is one national quantity that must be divided, not three quantities to be estimated separately.

Annual generation is

$$G \;=\; 365 \sum_{d} \left(P_d^{u} g_u + P_d^{r} g_r \right), \tag{1}$$

over districts d, with g_u and g_r the urban and rural per-capita rates. The collected fraction, which is what reaches a disposal site,

$$D \;=\; 365 \sum_{d} \left(P_d^{u} g_u (1-u_u) + P_d^{r} g_r (1-u_r) \right), \tag{2}$$

uses the uncollected fractions u_u and u_r, and the uncollected remainder is $U = G - D$ by construction rather than by a second estimate. Of that remainder, the burned quantity is $B = f_b U$.

The identity $D + U = G$ is asserted in code with exact equality, and a test fails if it ever ceases to hold. (Tier status is stated per component rather than per sector, in Appendix E: the categories share one country-specific activity basis but differ in how much of each calculation is a default.) This is the property that separate estimation cannot provide. If landfill tonnage comes from one source and open-burning tonnage from another, their sum is unconstrained and can exceed or fall short of national generation without anything in the calculation noticing; here, an argument about the collection rate moves emissions between 4A and 4C2 and changes the total only through the different methane yields of the two fates.

For FY2023–24 this gives 31.56 Mt generated, 15.88 Mt collected to disposal, 15.67 Mt uncollected and 9.40 Mt openly burned. The partition is published as an executable fate ledger (`outputs/waste_ledger_2024.csv`, the same claimant structure as the transport fuel ledger), whose writer raises if either tier fails to close and whose final row is the fate this inventory does not model: 6.27 Mt of uncollected waste that does not burn—dumped on land or tipped into watercourses, carried explicitly rather than silently.

4.2. Solid-waste disposal: First Order Decay (4A)

Methane from a disposal site in year t depends on every year's deposit still decaying, so IPCC Volume 5 Chapter 3's first-order decay model is run over a deposit history rather than applied to a single year's tonnage. Decomposable degradable organic carbon accumulates and decays as

$$\mathrm{DDOCm}_a(t) = W(t)\,\mathrm{DOC}\cdot\mathrm{DOC_f}\cdot\mathrm{MCF} + \mathrm{DDOCm}_a(t-1)\,e^{-k}, \tag{3}$$

$$\mathrm{CH_4}(t) = \left[\mathrm{DDOCm}_a(t-1)\left(1-e^{-k}\right)\right] F \cdot \tfrac{16}{12}, \tag{4}$$

with $k = 0.065$, $\mathrm{DOC_f} = 0.5$, $\mathrm{MCF} = 0.6$ and $F = 0.5$.

Two parameters are deliberately zero. No methane recovery is credited, because none is documented at national scale and crediting an undocumented recovery would understate the category. The oxidation factor is zero for the same class of reason: IPCC's non-zero oxidation applies to managed sites with a cover layer, and Pakistani disposal is predominantly unmanaged, where the thin intermittent cover that drives oxidation is absent.

4.2.1. Spin-up, and why it is reported rather than assumed

The activity panel begins in 2000, but waste deposited before then is still emitting. Starting the model at the panel's first year with an empty site would assert that Pakistan landfilled nothing before 2000, which is false and biases the estimate low.

The deposit series is therefore back-projected to 1960 at 40 % of the panel's first-year tonnage, and the model is run from there. Because that back-projection is an assumption rather than data, the sector reports what it is worth: without spin-up the category would be 8.07 Mt instead of 9.17 Mt, so pre-2000 deposits contribute 12 % of today's landfill methane. A reader who prefers to start at the panel can read the alternative directly from the output file rather than recomputing it.

4.2.2. The monthly state engine, and why it is one recursion

The national release resolves the disposal stream into fiscal months, and the engine behind those months is this model

run at a monthly step rather than a reshaped annual answer: one continuous single-pool recursion from the 1960 spin-up, $S_m = S_{m-1}\, e^{-k\, d_m/365.25} + W_m$, where d_m is the month's exact calendar length (leap Februaries included) and W_m deposits the month's own calendar year's tonnage, days-weighted within the year. Running one recursion rather than a per-year replay matters for three named reasons: state flows across fiscal boundaries (June's stock is July's opening balance, so no seam appears at the year change); each month deposits its *own* year's waste (a per-year replay had injected the panel's maximum year into every year's stock); and the reaction delay between deposit and the onset of decay is a declared parameter (`swds_delay_months`, the IPCC six-month convention) applied as a shape refinement only. The single pool is kept deliberately — the annual model carries one decay constant, and a cohort structure would claim a waste-composition split no Pakistani record supports. The annual control is preserved exactly by construction: each fiscal year's twelve shares are normalised against its own annual figure, so the monthly engine decides only *when* within the year, never how much, and the within-year spread it computes is printed into the month rows' basis rather than asserted in prose.

4.3. Open burning (4C2)

IPCC Equation 5.1 applied to the burned quantity B, taking that chapter's own allowance to combine the dry-matter, carbon-content and fossil-fraction terms into one coefficient rather than carrying them separately:

$$\mathrm{CO_2} = B \cdot \mathrm{CF_{fossil}} \cdot \mathrm{OF} \cdot \tfrac{44}{12}, \tag{5}$$

where $\mathrm{CF_{fossil}}$ is the fossil carbon per *wet* tonne given in Section 3 and the oxidation factor is $\mathrm{OF} = 0.71$. Dry matter does not appear here because it is already inside that coefficient; it enters this category only through the N_2O factor, which IPCC states per dry tonne. The burned quantity B is itself IPCC Equation 5.7.

Only the fossil carbon fraction produces reportable CO_2. The biogenic fraction is reported as a memo item at 3520 Gg CO_2, outside the national total, because that carbon was fixed from the atmosphere within the year. Methane and nitrous oxide from the same burning *are* counted regardless of carbon origin: incomplete combustion is a net source however the carbon entered the waste stream.

4.4. Domestic wastewater (4D1)

Tier 1 over the sanitation-resolved population. Total organically degradable material is

$$\mathrm{TOW} = P \cdot \mathrm{BOD} \cdot 365 \cdot I, \tag{6}$$

with $I = 1.25$ the industrial-and-commercial correction, and methane is

$$\mathrm{CH_4} = \sum_j \mathrm{TOW}_j \cdot B_0 \cdot \mathrm{MCF}_j, \tag{7}$$

summed over sanitation pathways j with $B_0 = 0.6$ kg CH_4 per kilogram BOD. The pathway correction factors are the subject of Section 9.

Nitrous oxide is computed from protein consumption rather than from BOD, by the method of the 2019 Refinement (Volume 5 Chapter 6, Equations 6.7 to 6.10a) (IPCC, 2019). Protein supply is reduced to protein consumed (a factor of 0.96) and its nitrogen (0.16) is allocated to the same sanitation pathways the methane uses, with factors of 1.1 for household products, 1.1 for non-consumed protein (the chapter's updated Table 6.11, taken over its text, which gives no increment where food waste goes with solid waste) and, for industrial and commercial co-discharge, 1.25 on the sewered population only. Treatment removes part of each pathway's nitrogen before discharge—0.15 in septic tanks, 0.12 in latrines and 0.40 in the treated share of sewered flow, which is declared at 1 % and not measured—and the nitrogen entering a treatment plant emits there at 0.016. What is left is summed over every pathway and emits at the effluent factor of 0.005; the whole population stays in the estimate. The result is 9.71 Gg of N_2O. The single equation of the 2006 Guidelines, which this inventory used before and which treats one nitrogen total on protein supply as effluent, is retained as the alternative and gives 11.45 Gg. Neither method reads BOD or the sewer correction factor. This is why nitrous oxide does not move under either of the sensitivities reported below, and why a sensitivity row that reported methane alone would be misread as a category total—a distinction this sector's output file previously got wrong and now states explicitly.

5. Emission Factors and Parameters

Every factor and parameter used here is an IPCC 2006 default, a 2019 Refinement default (the wastewater nitrous oxide chain) or a documented regional value, with its source recorded in the versioned parameter table. Three deserve discussion because the choice was not automatic.

5.1. Two oxidation factors that must not be reconciled

This sector carries an oxidation factor of 0.71 for open burning and 0 for landfill, while the fossil-fuel sectors of the same inventory carry none at all. That looks like an inconsistency and is not.

The 2006 Guidelines fold complete oxidation into the fuel combustion emission factors themselves—Table 1.4 carries an explicit oxidation column of 1.0 for every fuel—so applying the 1996-vintage 0.98 on top of a 2006 factor double-discounts. Open burning is different in kind: it genuinely is incomplete, and 0.71 is the 2006 value that says so. Landfill oxidation is different again, describing methane oxidised while passing through a cover layer, which unmanaged Pakistani sites largely do not have.

Three different numbers describing three different physical situations. Tests pin all three, so that a future reader who notices the apparent inconsistency and "corrects" it fails the build rather than silently changing the sector.

Parameter	Applies to	Value
Generation rate, urban	4A, 4C2	0.453 kg/cap/day
Generation rate, rural	4A, 4C2	0.283 kg/cap/day
Uncollected fraction, urban	4A, 4C2	0.30
Uncollected fraction, rural	4A, 4C2	0.70
Burned fraction of uncollected	4C2	0.60
Decay constant k	4A	0.065 yr^{-1}
DOC_f	4A	0.50
MCF (site management)	4A	0.60
F (CH_4 in landfill gas)	4A	0.50
Oxidation factor	4A	0
Methane recovery	4A	0
Back-projection to 1960	4A	40% of first panel year
Dry matter	4C2	0.70
Total carbon	4C2	0.2195 t C/t wet
Fossil carbon	4C2	0.0757 t C/t wet
Oxidation factor	4C2	0.71
BOD	4D1	40 g/cap/day
B_0	4D1	0.6 kg CH_4/kg BOD
MCF, sewered / septic	4D1	0.50
MCF, latrine / open	4D1	0.10 / 0
Industrial-commercial correction	4D1	1.25

Table 1: Parameters, with the IPCC volume and table recorded against each in `data/waste_params.csv`. The two zeros in the 4A block are deliberate and argued in Section 4, not defaults left unset. Full row-level provenance—value, uncertainty, source and basis for every parameter and factor—is the generated table in Appendix F.

5.2. *Fossil carbon, and why only a third of the burning CO_2 is reported*

Total carbon in Pakistani municipal waste is 0.2195 t per wet tonne, of which 0.0757 t—roughly a third—is fossil. Only that third produces reportable CO_2; the rest was fixed from the atmosphere by plants within the year and is reported as a memo item at 3520 Gg.

This convention is easy to apply inconsistently, because the same waste produces CH_4 and N_2O that *are* counted in full regardless of carbon origin. The distinction is not about the waste but about the atmosphere: biogenic CO_2 returns carbon that was recently removed, while methane from the same carbon is a different molecule with a different radiative effect and no such offset.

5.3. *What the parameters are worth*

Three of the parameters above dominate the sector's uncertainty, and each is examined as a published sensitivity rather than left implicit: the sewer MCF (Section 9), the BOD rate, and the spin-up back-projection. The generation and collection rates are not among them—not because they are well known, but because the partition means they move emissions between categories more than they move the total.

6. Validation

There is nothing to validate this sector against directly. No Pakistani facility reports landfill gas, no drain is monitored for dissolved methane, and no survey measures how much uncollected waste burns. What exists is other people's models, and comparing a model against a model establishes consistency rather than accuracy. This section says what that comparison does and does not show.

6.1. *Against EDGAR*

EDGAR reports Pakistani waste—disposal, burning and wastewater together—at 26.57 Mt CO_2e for 2023, year-matched to this inventory's reference year (Crippa et al., 2025). This inventory reports 37.45 Mt, or 141 % of it. (EDGAR's 2024 value, 27.21 Mt, is carried in the benchmark file for release-vintage context and is not the comparison basis.)

That row has changed sides during this work, which is what makes it informative. Before domestic wastewater was built, this sector covered disposal and burning only and read 49 % of EDGAR; the gap was then a measure of what the missing category would add. With 4D1 in, the sector overshoots. The same comparison has therefore diagnosed an omission and then a possible overestimate, which is more than a single ratio usually earns.

The likeliest single cause is the sewer methane correction factor. At IPCC's stagnant-sewer 0.5 the wastewater category is 24.46 Mt and the sector overshoots; at the river-discharge 0.1 it is 15.39 Mt, the sector is 28.38 Mt, and the comparison lands at 1.07 times EDGAR. No other single parameter in the sector can move it that far.

It would be easy to treat that as confirmation and adopt 0.1. We do not, and Section 9 argues why: EDGAR is itself a model making its own assumption about the same unmeasured quantity, so agreement between the two would demonstrate that two models had made the same choice rather than that the choice is right.

6.2. *Against Climate TRACE*

Climate TRACE's Pakistani waste subsectors are built from a different evidence base again, including facility detection. Its solid-waste figures are broadly consistent with the disposal category reported here; its coverage of open burning and domestic wastewater is not constructed on a comparable scope, so a total-to-total comparison would mislead rather than inform and is not reported as one.

6.3. *Against household survey evidence*

The 2024–25 Household Integrated Economic Survey asks every sampled household how its waste is collected or disposed of (Pakistan Bureau of Statistics, 2025), and its design-weighted answers are the first Pakistani survey evidence bearing on the partition's load-bearing parameters. They disagree with the What a Waste collection coverage the partition is built on, in one direction: 42.0 % of urban households report no collection of any kind against the 30 % assumed, and 93.4 % of rural households against 70 %. The route detail supports the open-dumping reading of the uncollected stream—88.5 % of rural households report disposal to open space, and under 1 % to waterways.

Two caveats keep this a comparison rather than a correction. Household route shares are not waste-mass shares: the survey weights households, not tonnes, and converting one to the other needs stratum generation rates it does not carry. And counting public-bin use as collected overstates collection where bins are cleared infrequently, which the same module records. The configuration is therefore priced as a named structural scenario in the supplement rather than adopted; because the partition conserves generated waste, moving the uncollected fractions to the survey values moves emissions from slow landfill decay to immediate burning without changing the waste the country produces.

6.4. Internal checks that do have teeth

Three checks are enforced on every run and, unlike the external comparisons, can fail.

The mass balance closes. Disposed plus uncollected equals generated, exactly. This cannot detect a wrong collection rate, but it can and does detect the class of error where a change to one category silently fails to appear in the other.

The burned quantity is exactly the burned fraction of the uncollected remainder. Asserted with equality against an independently recomputed value, so a future refactor that accidentally applies the burned fraction to total generation rather than to the uncollected part fails immediately. That specific error would inflate open burning by a factor of two and is entirely plausible.

Provincial allocation reconciles exactly. The four provinces and the territories sum to the national figure for each category, not merely for the sector.

6.5. What would constitute real validation

Three measurements, in order of what they would settle. Dissolved-methane and flux sampling on urban drains across a seasonal cycle would resolve the correction factor that dominates the sector. Landfill gas monitoring at two or three large sites would test the first-order-decay parameters directly. A waste-characterisation survey at the same sites would replace the regional composition figures with Pakistani ones and firm up the fossil carbon fraction that governs reportable CO_2 from burning.

None is expensive relative to what the sector contributes to national uncertainty. That argument is made quantitatively in Section 9.

7. Results

7.1. The sector

Pakistan's waste sector emitted 37.45 Mt CO_2e in FY2023–24 on AR5 metrics, or 36.37 Mt on AR6. Domestic wastewater is 65.3 % of it.

Methane is 87 % of the sector, which is why the AR5-to-AR6 difference of 1.08 Mt is almost entirely a methane effect and why this sector is unusually sensitive to the metric chosen.

Category	Gas	Gg	Mt CO_2e	±%
Open burning (4C2)	CO_2 (fossil)	1853.21	1.853	84.3
	CH_4	61.12	1.712	92.9
	N_2O	0.99	0.262	124.2
Solid-waste disposal (4A)	CH_4	327.45	9.169	79.8
Domestic wastewater (4D1)	CH_4	781.63	21.886	73.7
	N_2O	9.71	2.572	268.0
Total			**37.453**	**51.1**

Table 2: Waste-sector emissions, FY2023–24, AR5 GWP-100. Biogenic CO_2 from the biomass content of burned waste, 3520 Gg, is a memo item outside the total per Vol 5 Ch 5 §5.1. The sector band is the Approach-1 quadrature over these six components.

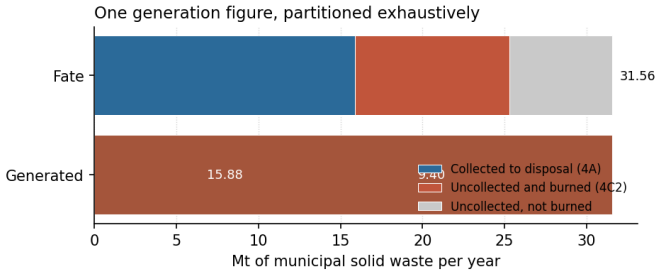


Figure 1: The partition. 31.56 Mt of municipal solid waste is generated; 15.88 Mt is collected and enters the decay model as 4A; 15.67 Mt is uncollected, of which 9.40 Mt burns as 4C2 and the remainder is dumped without combustion. The two lower bars sum to the upper one exactly, which is the property the code asserts rather than checks.

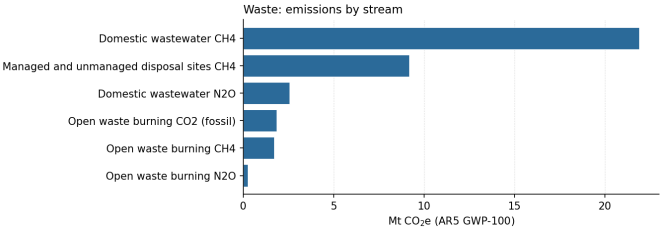


Figure 2: Emissions by stream. Domestic wastewater methane alone, at 21.89 Mt CO_2e, is larger than the other five streams combined.

7.2. Where the uncertainty is

Five of the six streams carry bands between 74 % and 124 %; the sixth, wastewater N_2O, declares 268.0 %—the log-variance-matched equivalent of its factor's bounded 150-fold published range, repriced after the previous symmetric 100 % was shown incompatible with the stated evidence. Together they give ±51.1 % for the sector under Approach 1—the widest of any major sector in the national inventory this sector belongs to; only domestic aviation, at 0.1 % of the national total, carries a wider band (±58.8 %).

The bands are wide because the activity data are modelled rather than measured, and the Approach-1 figure does not carry their shape: the sector's own parameter-rerun treatment—inputs drawn, the engine rebuilt and the fate ledger asserted in every one of 10 000 realisations—puts the 95% interval at 25.7 Mt to 61.4 Mt, asymmetric by construction, and the NATIONAL Monte Carlo consumes those same draws, printing ±46.4 % as this module's symmetric summary where

the retired lognormal surrogate said 58.2 % (Supplementary Note S1). A symmetric percentage cannot represent that shape; it is quoted here only because the quadrature convention requires one number.

7.3. Sensitivities, published rather than chosen

Assumption	Category (Mt)	Sector (Mt)
Sewer MCF 0.5 (reported)	24.46	37.45
Sewer MCF 0.1 (river reading)	15.39	28.38
BOD at India's 34 g/cap/day	21.17	34.17
Disposal with spin-up (reported)	9.17	37.45
Disposal without spin-up	8.07	36.35

Table 3: The three assumptions this sector cannot settle, each reported as an alternative rather than absorbed into the central estimate. Nitrous oxide is invariant under both wastewater cases because it derives from protein rather than from BOD or sewer condition, so all rows are category totals including it—a distinction this sector's own output file previously got wrong.

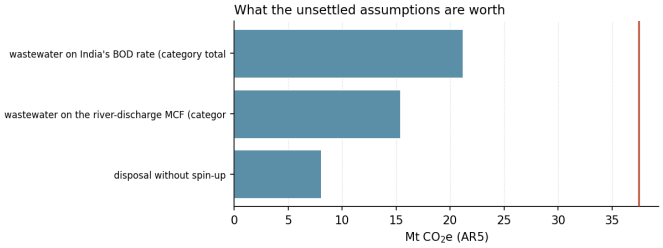


Figure 3: The same result as a range around the reported figure. A reader who has to reconstruct a range from prose will not reconstruct it.

The spin-up row answers a question most first-order-decay implementations answer implicitly: pre-2000 deposits contribute 12 % of today's landfill methane. An implementation that starts at its activity panel is not making no assumption; it is asserting that the country landfilled nothing before its data begin.

The supplement extends these point sensitivities to named structural scenarios with conditional intervals (Table 8), including a mixture representation of the sewer correction factor—stagnant, flowing and treated regimes weighted explicitly—and the partition on the HIES household route shares of Section 6.

7.4. Provincial allocation

The provincial pattern follows population closely, which is expected for a sector whose activity datum *is* population, and is the reason this sector's geography is less informative than, say, the power sector's. What the allocation locates is the *origin* of the waste and wastewater—where the generating people live—not the release: disposal methane is emitted at landfills and dumps, wastewater methane along drains, at treatment works and at discharge points, and burning at burn sites. What the inventory holds toward that release geography is a prospective seed only: 56 OpenStreetMap `landuse=landfill` polygons (OpenStreetMap contributors, 2026) (ODbL 1.0 derivative database, `data/gis/landfills/`), 13 of them named,

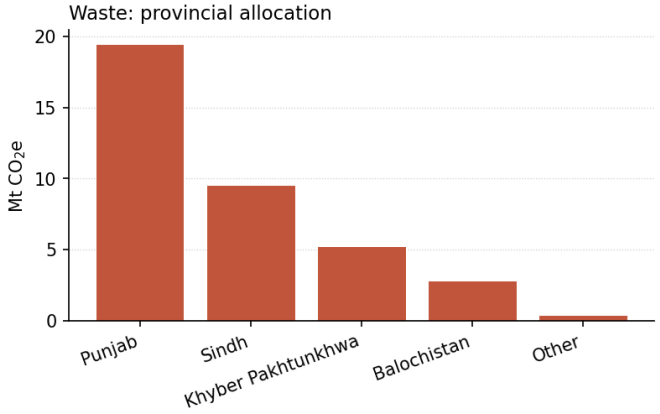


Figure 4: Punjab 19.49 Mt, Sindh 9.55 Mt, Khyber Pakhtunkhwa 5.21 Mt, Balochistan 2.80 Mt and 0.40 Mt in the territories. Allocated on the same district population panel that drives generation. This is an *origin* allocation: population locates where waste and wastewater are produced, and the split reconciles exactly against each category's national figure—but it carries the further assumption that release is co-located with origin, which is the stronger claim.

quality-flagged, and touched by no estimated stream—far too sparse for an SWDS allocation, and registered so the release can cite what it holds rather than claiming it holds nothing. The three objects are published SEPARATELY, as the review asked: origin (`waste_district_2024.csv`), fate (the ledger), and release—the polygons republished as candidates (`outputs/waste_release_candidates_2024.csv`) with an assignment product (`outputs/waste_release_assignment_2024.csv`) whose defining column is EXPLICIT UNASSIGNED MASS: a district's disposal emissions are assigned only to its own mapped candidates (equal split, capacity unknown and declared), nothing crosses a district boundary, and 75 % of the disposal category stays unassigned across 113 districts—the honest measure of how far a release map is from existing. At provincial resolution the origin/release distinction is modest, because waste rarely crosses a provincial boundary between generation and fate; at any finer resolution the population surface would place landfill methane in residential neighbourhoods, which is why the sector's ALLOCATION remains origin-based and the limitations section carries the gap.

7.5. Contribution to the national inventory

Within the ten-sector national inventory this sector belongs to, its three categories rank very differently by size and by contribution to national uncertainty:

Category	Level (%)	Variance (%)
Domestic wastewater	5.5	13.0
Solid-waste disposal	2.06	2.27
Open burning	0.86	0.22

Table 4: Shares of the national total and of national Approach-1 variance. Domestic wastewater contributes more than twice as much uncertainty as emissions; open burning contributes a quarter as much.

On that Approach-1 screen domestic wastewater is the second largest contributor to national inventory variance of any category in the country, behind only enteric fermentation. That is the quantitative form of the argument in Section 9: the measurement that would settle this sector's central assumption is worth more to the national inventory than its size alone suggests.

8. Air-Pollutant Co-Emissions

Open burning of municipal waste is a combustion source in the middle of where people live, so its air-pollutant emissions matter for reasons its greenhouse-gas emissions do not—though the burn sites themselves are unmapped, so its geography, like the sector's, is an origin allocation on population rather than observed release locations. This section reports them on the same activity basis as Section 7: the same 9.40 Mt of burned waste, through the Andreae (2019) garbage-burning factors, reported as mass and never converted to CO_2e.

Pollutant	kt/yr
CO	434.45
NMVOC	151.40
$PM_{2.5}$	63.85
HCl	18.43
NO_x (as NO_2)	15.14
Black carbon	9.22
NH_3	5.92
SO_2	3.29

Table 5: Air pollutants from open waste burning, FY2023–24. Mass only; these never enter a greenhouse-gas total. Computed from the same burned tonnage as the 4C2 emissions above, so the two describe the same fire. NO_x is reported as NO_2-equivalent since August 2026 (Andreae publishes the garbage-burning factor as NO; the conversion is documented on the factor row), the inventory-wide basis.

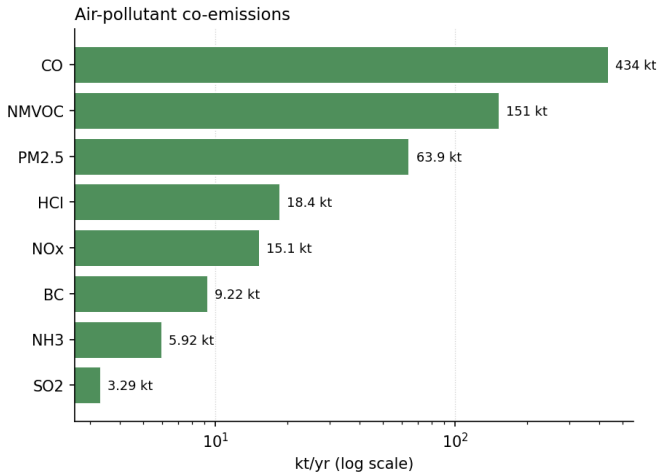


Figure 5: The same figures on a log axis, which the range demands: carbon monoxide and sulphur dioxide differ by more than two orders of magnitude, and a linear axis renders everything below $PM_{2.5}$ as a line of zero width.

8.1. Where this sits among Pakistani particulate sources

At 63.85 kt of primary $PM_{2.5}$, open waste burning is the second largest source in the national inventory that contains it—behind residential solid fuel at 282 kt and ahead of crop-residue burning at 54 kt and road transport at 8.2 kt (EMEP/EEA Tier 1 since the air program's factor refactor).

That ranking is worth stating carefully, because it inverts the greenhouse-gas one. Open burning is 0.86 % of national CO_2e and roughly 14 % of national primary $PM_{2.5}$: a category that is nearly negligible for climate and substantial for exposure. Road transport runs the other way, at 9.03 % of emissions and a twentieth of the particulate.

Because every sector's air layer in this inventory is computed from that sector's own greenhouse-gas activity data, the comparison is a statement about how these sources burn rather than an artefact of four studies using four different activity bases—which is the usual reason such comparisons are unreliable in developing-country settings.

8.2. Black carbon and hydrogen chloride

Two entries deserve comment because they are specific to waste rather than generic to combustion.

Black carbon at 9.22 kt is a strong short-lived warming agent that this inventory reports as mass and does not convert to CO_2e, following the convention that its metrics remain contested. It is reported because omitting it would understate the climate relevance of open burning even while the greenhouse-gas total for 4C2 is small.

Hydrogen chloride at 18.43 kt appears in no other sector of this inventory and is a direct consequence of burning chlorinated plastics in the open. It is a marker for the same combustion conditions that produce dioxins and furans, which this inventory does not estimate and which would need a dedicated study; its presence here is a flag for that gap rather than a complete account of it.

8.3. Limitations of this layer

The burned tonnage carries the full uncertainty of the partition, so every figure above inherits it. The composition is regional rather than Pakistani, which matters more for pollutants than for greenhouse gases because plastics and textiles drive the chlorine and much of the particulate. And the factors describe open burning generically rather than the specific conditions of Pakistani street-side burning—smouldering, damp, and often mixed with organic waste—which would raise the incomplete-combustion products relative to these values.

9. Discussion

9.1. The methane correction factor, and why it is not resolved here

The sewer methane correction factor is the largest single assumption in this sector and the one the evidence does not settle. It deserves the space because how it is handled is a methodological position, not a parameter choice.

IPCC gives 0.5 for stagnant open sewers and 0.1 for wastewater discharged to a flowing, well-aerated body. The distinction was drawn for systems that are clearly one or the other. Pakistani urban drainage is not: the drains are open, warm, slow-moving and heavily organically loaded, which is the description 0.5 exists for, while a substantial fraction ultimately discharges to rivers and irrigation channels, which is the description 0.1 exists for. Both descriptions are accurate about different parts of the same system, and no Pakistani source reports the hydraulic condition of the sewered fraction, which is the quantity the factor is actually asking about.

The consequence is a factor of 1.59 on the sector's largest category: 24.46 Mt against 15.39 Mt, or 37.45 Mt against 28.38 Mt for the sector.

9.1.1. Why external agreement is not a tiebreak

At 0.1 this sector lands close to EDGAR, and it would be easy to present that as corroboration. It is not, and treating it as one would be a methodological error worth naming because it is a tempting one.

EDGAR is a model. Its Pakistani wastewater figure rests on its own assumption about the same unmeasured quantity—the anaerobic fraction of Pakistani sewage—made from the same absence of Pakistani measurement. If this inventory adopted 0.1 because it produces agreement, the two would agree because both had chosen the same default, and the agreement would then be cited as though it were evidence about Pakistani drains. That is how a shared assumption becomes an apparent consensus.

We therefore report 0.5 as central, because the physical description of the drains is the stronger evidence and is specific to Pakistan, and publish 0.1 beside it so a user who weighs the comparator differently can take it. What we do not do is let the comparison choose.

9.1.2. What would settle it

Dissolved-methane concentration and flux measurement on urban drains in two or three Pakistani cities, sampled across a seasonal cycle to capture the dry-season stagnation and the monsoon flushing that the two IPCC categories describe separately.

The case for doing it is quantitative rather than rhetorical. Domestic wastewater is 5.5 % of Pakistan's national emissions and 13.0 % of national inventory variance under Approach 1—the second largest variance contributor of any category in the country on that screen. A measurement campaign of a few tens of thousands of dollars would reduce national inventory uncertainty more than any comparable expenditure elsewhere in the waste or energy sectors.

9.2. What the closed mass balance buys, and what it does not

The balance guarantees that the collected and uncollected streams together account for exactly the waste the country generates, with the burned quantity a declared fraction of the uncollected stream and the unburned 6.27 Mt carried as an explicit unresolved fate rather than assigned to either category. That removes a specific class of error—the two categories drifting into mutual inconsistency as one is updated—and makes the collection rate an explicit, single, arguable number rather than an implicit consequence of two independent estimates.

It is worth being precise about what it does not buy, because a closed balance invites overconfidence. It says nothing about whether the split point is right. The collection rate of 50.3 % carries real uncertainty, and moving it moves waste between two fates with quite different methane yields per tonne: a tonne landfilled decays slowly and anaerobically over decades, while a tonne burned emits immediately and mostly as CO_2. Nor does it validate the generation figure itself; a partition of a wrong total still closes.

What it changes is the shape of the argument. A reader who disagrees with the collection rate can see exactly which two categories move and in which directions, rather than having to work out whether two independently sourced numbers are mutually consistent.

9.3. Implications

Three follow, in descending order of confidence.

Wastewater is the sector, and the sector is a measurement problem. Two-thirds of the emissions and nearly all of the tractable uncertainty sit in one category whose central parameter has never been measured in Pakistan. Effort spent refining the solid-waste decay parameters or the burning composition would improve smaller numbers that are already better constrained.

Open burning matters for air quality, not for climate. At 0.86 % of national CO_2e and roughly 14 % of national primary $PM_{2.5}$, it is a category whose case for intervention rests on exposure. An abatement programme justified on greenhouse-gas grounds would be hard to defend; the same programme justified on particulate would be straightforward.

The collection rate is a policy variable, not just a parameter. Because the partition makes it the single number governing what fraction of waste burns rather than decays, improving collection is visible in this inventory as a movement between categories. That is the right structure for evaluating a municipal collection programme, and it is not available in an inventory whose two categories are estimated independently.

10. Limitations

The sewer methane correction factor is unresolved, and it is the sector. A factor of 1.59 on two-thirds of the emissions, argued in Section 9. Both readings are published; neither is measured.

Industrial solid waste and industrial wastewater are not estimated at all. No national quantity for either has been compiled here. The categories reported here are municipal and domestic only, and both gaps carry the notation key NE with that justification, in the complete sector-4 matrix of Appendix G—which also declares biological treatment (4B) and incineration (4C1) as NE rather than leaving them silently absent. Industrial wastewater is the more consequential omission: given that domestic wastewater alone is 5.5 % of the national inventory, an

industrial stream from a textile and tanning sector of Pakistan's size is unlikely to be negligible.

Generation rates are single national values. The urban and rural per-capita rates do not vary by province or city size, which is a real simplification for a country whose largest city and smallest districts differ substantially in consumption and in what they discard. The rates are regional literature values rather than Pakistani measurements.

The collection rate is the softest number in the partition. The mass balance closes by construction, but 50.3 % is an assumption, and it governs how much waste decays slowly in a landfill versus burns immediately in the street. It is a single national figure applied through an urban/rural split, when collection in practice varies between municipalities far more than between those two categories. The one Pakistani survey that asks the question directly, HIES 2024–25, reports materially less collection than the assumption (Section 6); its household-route shares are priced as a structural scenario rather than adopted, because route shares are not mass shares.

The burned fraction is a literature value. That 60 % of uncollected waste burns is the least evidenced parameter in the sector and has no Pakistani measurement behind it at all. It scales open burning linearly, and with it the entire air-pollutant layer in Section 8. Its complement is the fate ledger's unresolved row: 6.27 Mt of uncollected, unburned waste—dumped to land and waterways—to which no emissions pathway is applied at all. That choice is now bounded rather than silent: the scenario table prices the unresolved mass as unmanaged shallow disposal (sector total 39.87 Mt) and unmanaged deep disposal (42.28 Mt) against the reference reading (37.45 Mt), on the declared assumption that the dumped stream shares the collected stream's deposit history and composition. Waterway decay lies between inert and shallow. No probability attaches to any point in the bound—a disposal-site census, not a better default, is what would close it.

Waste composition is regional, not Pakistani. The fossil carbon fraction governs how much of the burning CO_2 is reportable, and the plastics and textiles content governs the chlorine and much of the particulate. A Pakistani waste-characterisation survey would replace both.

No methane recovery is credited and no landfill oxidation applied. Both are set to zero because neither is documented at national scale, and both choices bias the disposal category high if any large Pakistani site does in fact manage its gas or maintain a cover. The direction of that bias is stated rather than hedged.

The spin-up back-projection is an assumption, not data. Extending the deposit series to 1960 at 40 % of the panel's first year is a judgement about Pakistan's waste history. Its effect is reported—12 % of current landfill methane—so a reader can remove it, but the shape of the back-projection is not evidenced.

Uncertainties are large, and honestly so—and honestly asymmetric. Five of the six streams carry between 74 % and 124 %; wastewater N_2O carries the bounded 150-fold factor range, and the sector reads ±51.1 % by quadrature with a parameter-rerun 95% interval of 25.7 Mt to 61.4 Mt (Supplementary Note S1). These follow IPCC's own Tier 1 guidance for waste parameters and reflect that the underlying activity is modelled from population rather than measured at facilities. They are not an artefact of conservative parameter choice; they are what Tier 1 can say about a country with no facility-level waste data.

Dioxins and furans are not estimated. Open burning of chlorinated plastics produces them, and the hydrogen chloride reported in Section 8 is a marker for the conditions that do. Estimating them needs a dedicated study rather than a factor applied to the same tonnage.

11. Conclusion

Pakistan's waste sector emitted 37.45 Mt CO_2e in FY2023–24 on AR5 metrics: 24.46 Mt from domestic wastewater, 9.17 Mt from solid-waste disposal and 3.83 Mt from open burning. The sector is 8.4 % of a ten-sector national inventory and 87 % methane, which makes it unusually sensitive to the metric chosen and unusually relevant to near-term mitigation.

Two things generalise beyond this country.

An activity figure shared between categories should be partitioned, not estimated twice. Solid-waste disposal and open burning describe two fates of one waste stream, and computing them from one generation figure with $D + U = G$ asserted removes a class of inconsistency that independent estimation cannot detect. The cost is nil and the gain is that an argument about the collection rate becomes an argument about where emissions sit rather than about how much waste exists. The same shape recurs wherever two categories divide one national quantity.

Where an assumption spans a factor approaching two and the evidence does not choose, the defensible output is both numbers. The sewer methane correction factor moves this sector from 37.45 Mt to 28.38 Mt. The lower value happens to agree better with EDGAR, and adopting it for that reason would have been a mistake: EDGAR is a model resting on its own assumption about the same unmeasured quantity, so the agreement would have recorded a shared default as though it were evidence about Pakistani drains. We report 0.5 as central because the physical description of the drains is the stronger and more specific evidence, publish 0.1 alongside, and name what would settle it.

For Pakistan specifically, the analysis produces one clear recommendation. Domestic wastewater is 5.5 % of national emissions and 13.0 % of national inventory variance under Approach 1—the second largest variance contributor of any category in the country on that screen, behind enteric fermentation. Dissolved-methane and flux sampling on urban drains in two or three cities, across a seasonal cycle, would reduce national inventory uncertainty more per unit of cost than any comparable measurement in the waste or energy sectors. It is a small campaign answering a question that currently has no Pakistani answer at all.

Everything reported here is reproducible. The engine, the activity data, the parameter tables and every output quoted are

public; each figure is regenerated from the outputs it plots and a test fails if any is stale; and the numbers in this manuscript are checked against the generated files they came from on every commit.

Code and data availability

The calculation engine, the activity data, the parameter tables and every output quoted here are version-controlled in the inventory repository (https://github.com/RehanAhmad13/pakistan-emissions-sectors) and reproduce from a single command. The sector is regenerated with `python3 -m pkinv.sectors.waste.run_waste`; Appendix A gives the full sequence. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

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Appendix A. Reproduction

The sector runs on Python 3.11 with no compiled dependencies:

```
pip install -e .
python3 -m pkinv.sectors.waste.run_waste
python3 -m pkinv.sectors.waste.air_pollutants
python3 -m pkinv.sectors.waste.uncertainty
```

which writes `waste_totals_2024.csv`, `waste_by_stream_2024.csv`, `waste_provincial_2024.csv`, `waste_air_pollutants_2024.csv`, `waste_ledger_2024.csv`, the twelve-month layer (`waste_monthly_2024.csv`, each month's share with the emissions it implies and the basis of the shape), the three `waste_uncertainty_*.csv` files and `waste_report.md` to `outputs/`. The generated appendix tables are rendered by the three `scripts/make_waste_*_table.py` generators, and a test asserts each committed table matches a fresh regeneration. The cross-sector layer that places this sector in the national inventory is:

```
python3 -m pkinv.report.national_rollup
python3 -m pkinv.report.key_categories_all
python3 -m pkinv.report.uncertainty_all
```

Figures are regenerated with `python3 -m pkinv.report.figures` and copied into this manuscript by `python3 scripts/sync_figures.py`; a test fails if any shipped figure differs from the output it plots. The full suite, including the sector's own arithmetic tests and the checks that every number printed above matches its source CSV, runs with `python3 -m unittest discover tests`.

Appendix B. Sources

Population is the 2023 Population and Housing Census (Pakistan Bureau of Statistics, 2023), interpolated to a district panel. Waste generation rates and composition derive from the Pak-EPA eight-city survey (Pakistan Environmental Protection Agency and Environmental Protection and Management Consultants, 1996), which Kaza et al. (2018) re-bins; collection coverage is the Pakistan row of Kaza et al. (2018). Sanitation shares are the WHO/UNICEF Joint Monitoring Programme's Pakistani series (WHO/UNICEF Joint Monitoring Programme, 2023). Emission factors and model parameters are IPCC 2006 Guidelines Volume 5 defaults (IPCC, 2006), with the wastewater nitrous oxide method and its factors from the 2019 Refinement (IPCC, 2019); air-pollutant factors are the Andreae (2019) garbage-burning assessment. External comparisons are EDGAR's year-matched series (Crippa et al., 2025) and Climate TRACE (Climate TRACE, 2026).

Every factor in the repository cites its source table, and a provenance test asserts that none lacks a citation and that any informally sourced value is explicitly declared as such.

Appendix C. Supplementary Note: the two sensitivity conventions

Both wastewater sensitivities move methane only. Nitrous oxide in 4D1 is computed from protein consumption rather than from BOD, so it is invariant under a change to either the sewer correction factor or the BOD rate.

This is worth an appendix because getting it wrong is easy and was in fact got wrong here. The sector's output file originally wrote the two sensitivity rows as *methane only* into a column whose every other row was a category total. A reader comparing the reported 24.46 Mt against the river-discharge row's 12.82 Mt would have computed a lever of 1.91 where the real one is 1.59, and the file disagreed with the module's own printed summary, which had always added the nitrous oxide back.

The rows now carry category totals and are labelled as such. The general point is that a sensitivity is only interpretable against a stated basis, and a sensitivity table mixing bases within one column is worse than no sensitivity table, because it invites a subtraction that looks valid.

Appendix D. Supplementary Note S1: Uncertainty, Approach 1 against Approach 2

This note carries the sector's uncertainty treatment. An earlier edition's form was indefensible and was rebuilt: the effluent N_2O factor carried a symmetric ±100 % against its own published 500-fold range (0.0005–0.25, the 2006 Guidelines'), three wastewater parameters carried zero uncertainty against stated ranges, and the Monte Carlo drew component totals that could not promise the fate ledger's closure in any single realisation. All three defects are repaired here, and every published band on this page moved as a consequence—which is the repair working, not a revision of the inventory (the repair left the deterministic totals unchanged to the last digit).

The bands moved a second time when the nitrous oxide estimate went onto the method of the 2019 Refinement (Section 4), and that move is a revision of the inventory: wastewater N_2O fell from 11.45 to 9.71 Gg and the sector from 37.91 to 37.45 Mt. The Refinement prints the effluent factor's range as 0.0005–0.075, a 150-fold range, and that is the range drawn below; the 2006 equation, drawn over its own range, is kept as the alternative.

Appendix D.1. The headline interval: a parameter-rerun Monte Carlo, ledger-true by construction

The sector's published interval now comes from a Monte Carlo that draws the sector's *inputs*—generation rates, the collection partition, composition, decay and correction factors—and reruns the full engine for every one of 10 000 realisations, asserting the fate identities $D + U = G$ and $B + R = U$ inside each draw before it is accepted (`outputs/waste_parameter_mc_2024.csv`). Covariance is not modelled; it *emerges*: the shared generation figure moves disposal and burning together, the uncollected fraction moves mass between them with opposite signs, and the decay rate's nonlinearity through the deposit stock is exact rather than linearised.

The effluent N_2O factor is drawn from a bounded split-triangular in log space over its published range, with the IPCC default 0.005 as mode and median: the endpoints bound every draw, half the probability lies each side of the default, and

the density falls to zero at the published limits. Its 95% interval runs from about one sixth of the default to 8.2 times it—a 49-fold span, wide because the evidence is wide—against the 6.6-fold span the old symmetric declaration smuggled in.

Appendix D.2. The quadrature cross-checks, re-anchored

Under IPCC Approach 1—each component's relative uncertainty combined in quadrature—the sector reads ±51.1 %, against 47.8 before the repair; Approach 1 consumes the effluent factor's bounded shape as its log-variance-matched symmetric equivalent (about 267 %), the honest degradation for a surface carrying one number. An earlier national engine re-expressed this sector as component-lognormals at the width then declared, and its upper arm arrived nationally at 63.5 Mt, well below the rerun engine's on the 2006 range then drawn—a material compression of the upper tail. **The national engine consumes this sector's own rerun draws** (a persisted 100 000-draw file on seed 4421, whose first 10 000 draws are bit-identical to the rerun table above; hash in the release manifest), so the national rows below ARE the sector engine's quantiles.

Appendix D.3. Why the treatments differ, and what the difference says

The quadrature figure (±51.1 %) is the analytic limit of a symmetric, independent, moderate-uncertainty simulation—and the rerun engine is deliberately none of those, so its ±46.4 % symmetric summary (really −33.3 %/ + 59.5 % about the median) stands apart, and the divergence is the information.

Within categories the right-skew is now carried, not hidden: solid-waste disposal is ±79.8 % analytically and ±80.2 % in the injected simulation, its simulated median of 8.66 Mt below its mean of 9.15 Mt—a long right tail no symmetric half-width can represent. A reader taking 9.15 ± 79.8 % as symmetric would infer a lower bound near 1.8 Mt; the simulation puts the 2.5th percentile at 3.61 Mt and the 97.5th at 17.50 Mt. The true interval is narrower below and wider above—and for wastewater, whose factor evidence covers a hundred-and-fifty-fold published range, the bounded rerun treatment now states how much wider ALL THE WAY TO THE NATIONAL ROW (+85.8 % upper arm on the wastewater category).

Appendix D.4. The sector is unaffected by fossil-factor correlation

The national inventory this sector belongs to reports a second Monte-Carlo scenario in which the shared IPCC fuel-carbon factors are drawn as one correlated realisation across every fuel-burning sector, on the reasoning that there is one carbon content of the diesel a country burns and if the default is wrong it is wrong everywhere at once.

Waste barely moves: ±46.4 % independent against ±46.4 % correlated. That is the expected result and worth stating as an internal check rather than a finding. Only the CO_2 from open burning derives from a fossil-carbon factor at all, and it is 4.9 % of the sector; the methane that dominates comes from decay and anaerobic digestion, which share no factor with any combustion sector. A sector that moved materially under that scenario would indicate a factor being shared where it should not be.

Appendix D.5. What dominates, and what a measurement would buy

Decomposing the sector variance by category under Approach 1 gives domestic wastewater 13.0 % of *national* inventory variance, solid-waste disposal 2.3 % and open burning 0.2 %.

Two consequences follow. Effort on open burning cannot improve the national inventory materially however well it is done—its entire variance contribution is a fifth of one percent—though it remains the sector's most important category for air quality. And the wastewater band has TWO named drivers, not one. An earlier edition of this note claimed the sewer methane correction factor unquestionably dominated; that claim was an artefact of pricing the effluent N_2O factor at a symmetric ±100 % against the 500-fold range the 2006 Guidelines printed. What the effluent factor carries is now read from the released ensemble and not from a ranking of parameters: frozen at its default in a rerun of the waste engine on the ensemble's own draws, it accounts for 4.1 % of national variance and 4.8 % of the excess in the upper tail, where the wastewater category as a whole accounts for 9.0 % and 10.5 % (`outputs/pakemis_uncertainty_drivers.csv`). The sewer factor's structural reading (0.5 against 0.1, a factor of 1.59 on the CH_4 side) is a scenario and has no share in that decomposition. What a measurement would buy therefore splits by pathway: a dissolved-methane campaign on urban drains resolves the sewer weights, and an effluent-pathway N_2O measurement—which no Pakistani study provides—is what would collapse the factor range that now sets the sector's ceiling.

That is the quantitative case for the measurement campaign proposed in Section 9, and it is a stronger case than the sector's size alone would make.

Appendix D.6. The partition's couplings, and a grouped Approach 1

The pooled treatment above draws the six components independently, but two of the three categories are built from one generation figure and one collection coverage, and those couplings pull in opposite directions. The generation rates are shared *positively*: if Pakistanis generate more waste than the Pak-EPA rates say, disposal and burning are both underestimated at once. The uncollected fractions are shared *negatively*: a tonne not collected is a tonne unavailable to the decay model and available to burn, so the partition term partially cancels between 4A and 4C2. A grouped Approach 1 carrying both couplings analytically—shared terms linear with their signed elasticities, residuals in quadrature—gives ±51.6 % against the independent ±51.1 % (`outputs/waste_uncertainty_2024.csv`). The generation coupling outweighs the anticorrelation, so the pooled treatment slightly

Category	Det.	Mean	P2.5	P50	P97.5
Domestic wastewater	24.46	26.53	14.88	25.07	47.19
Solid-waste disposal	9.17	9.21	3.61	8.73	17.58
Open burning	3.83	3.82	1.65	3.54	7.54
Sector	**37.45**	**39.55**	**25.70**	**38.30**	**61.42**

Table 6: The parameter-rerun Monte Carlo, Mt CO_2e (AR5): 10 000 input draws on the published seed 4421 (the `seed` column of `outputs/waste_parameter_mc_2024.csv`), the sector rebuilt and the fate ledger asserted in every realisation. The interval is deliberately asymmetric—the mean sits above the deterministic value and the upper arm is longer than the lower—because the effluent factor's published range is asymmetric and is now priced rather than symmetrised. No symmetric ± percentage can represent this shape, which is why the quadrature figures below are reported as cross-checks rather than as the sector interval.

Category	Mean	P2.5	P97.5	±%
Domestic wastewater	26.51	14.77	46.57	63.4
Solid-waste disposal	9.15	3.61	17.50	80.2
Open burning	3.83	1.65	7.70	85.7
Sector	**39.48**	**25.47**	**60.93**	**46.4**

Table 7: This sector's rows in the NATIONAL Monte Carlo: the sector's own parameter-rerun draws, injected as a 100 000-realisation superset of the 10 000-draw table above (seed 4421; the small quantile differences against that table are draw count, bounded by the published convergence diagnostics). Mt CO_2e (AR5). The ±% column is the symmetric summary; the published CSV carries the asymmetric −/+ arms.

understates the sector band; the two effects are decomposed on the output row rather than netted silently.

Appendix D.7. Structural scenarios: the sewer as a mixture, the partition under survey evidence

The IPCC choice this sector cannot settle—stagnant-sewer 0.5 against river-discharge 0.1—is physically a statement about *weights*: Pakistani drains stagnate in the dry season and flush in the monsoon, so the real system is a mixture of regimes, not a single row of Table 6.3. Because the methane equation is linear in the correction factor, any mixture is exactly a scalar, and the configurations below are named structural scenarios with conditional intervals rather than a probability distribution over weights the country has never measured. Each row reruns the full sector (`outputs/waste_uncertainty_scenarios_2024.csv`; 10 000 draws on seed 4321—its own stream, so the pooled national stream's seed 42 is untouched; an earlier edition called this seed "private", which conflated stream separation with non-publication and made the scenario intervals irreproducible from the manuscript).

Two rows carry the new information. The equal mixture—half the sewered load stagnant, half flowing, with the treated sliver of 1 % at the not-well-managed plant factor—gives an effective correction factor of 0.300 and a sector of 32.92 Mt: the physically plausible seasonal reading sits closer to the central estimate than to the river reading. And the HIES configuration prices the survey evidence of Section 6: moving the partition to the household route shares *lowers* the sector to 35.56 Mt, because the extra uncollected tonnage burns immediately instead of decaying at the disposal correction factor—the direction a reader might not guess, and the reason the partition's couplings deserve the grouped treatment above.

Scenario	Mt	P2.5	P97.5
Reference (sewer 0.5)	37.45	22.58	60.71
All-stagnant mixture endpoint	37.45	22.45	60.26
All-flowing mixture endpoint	28.38	17.54	44.92
River MCF at its floor (0)	26.12	16.41	41.22
River MCF at its ceiling (0.2)	30.65	18.93	48.48
Equal stagnant/flowing mix	32.92	20.04	52.04
BOD at India's rate	34.17	20.84	54.60
Disposal without spin-up	36.35	21.75	58.61
HIES household route shares	35.56	21.78	57.72
Unresolved fate: inert (reference)	37.45	22.38	59.61
Unresolved fate: shallow dumps	39.87	24.19	64.13
Unresolved fate: deep dumps	42.28	25.61	66.91
Effluent EF at its floor (0.0005)	35.17	endpoint, no interval	
Effluent EF at its ceiling (0.075)	73.03	endpoint, no interval	

Table 8: Structural scenarios, sector totals, AR5. Deterministic value with the 95% interval of a conditional Monte Carlo (10 000 draws, seed 4321): each interval is conditional on its configuration, and no probability is assigned across rows. The all-stagnant endpoint is identical to the reference by construction, which is the proof that the mixture representation changes nothing; the all-flowing endpoint reproduces the published river-discharge sensitivity, and the two river-MCF rows price that reading's OWN published 0–0.2 range, which previously carried a silent zero. The three unresolved-fate rows are a FORMAL STRUCTURAL FAMILY (`unresolved_fate`, named in their scenario notes): they bound the fate ledger's 6.27 Mt no-pathway remainder between inert and unmanaged deep disposal, on the declared deposit-history assumption stated in the scenario notes. The reference member (inert) is an EVIDENCE-ABSENCE choice, not a physical claim—no site census says where the dumped mass sits—and the family members may not be quoted separately. The two effluent-EF rows state the discrete endpoints of the 150-fold published range the bounded draw prices continuously—at the ceiling, wastewater N_2O alone reaches about 38 Mt; on the 2006 Guidelines' ceiling of 0.25 it reached about 152 Mt, which is the arithmetic an external reader used to show the old ±100 % was incompatible with the stated evidence. The endpoints are deliberately published without conditional intervals: an interval around a deliberately extreme point would manufacture spread the evidence does not license.

Five further rows of the same file price what the nitrous oxide method declares and does not measure. The equation of the 2006 Guidelines, the reference before, gives a sector of 37.91 Mt. Reading the Refinement's Figure 6.1 literally—the nitrogen leaving septic tanks, latrines, unimproved facilities and open defecation goes to soil, so that only sewered nitrogen is effluent—gives 35.87 Mt; the reference follows Equation 6.8, which sums every pathway. The Refinement's factor for nutrient-impacted or hypoxic receiving waters (0.019) on every discharge gives 44.57 Mt, an endpoint without an interval, because no classification of Pakistan's receiving waters is compiled. Septic tanks with a land dispersal field give 37.72 Mt, and twice the declared treated share of sewered flow gives 37.48 Mt. The two methods are also compared on

the same draws (`outputs/pakemis_wastewater_method_comparison.csv`): the wastewater category's 95% interval is 14.9 Mt to 47.2 Mt under the 2019 method and 15.1 Mt to 85.2 Mt under the 2006 method drawn over its own range, and the national interval is 399.8 Mt to 505.4 Mt against 401.0 Mt to 524.4 Mt.

Appendix E. Supplementary Table: tier status by component

Tier status differs by component. The three categories share one activity basis – the district population panel – but carry different levels of country-specific characterisation, and two pieces of the sector are not on the IPCC tier ladder at all. One word for the whole sector would either overstate the factors or understate the activity.

Appendix F. Supplementary Table: full factor and parameter provenance

Every parameter and factor the sector uses, with value, declared 95% uncertainty, PRICED BOUNDS where the source publishes a range, and abridged source, rendered from the versioned data files the engine reads. Note the deliberate basis differences the compact table in the body cannot carry: burning CH_4 is per WET tonne and N_2O per DRY tonne; the NO_x factor is converted to NO_2-equivalent ($\times 46/30$) from Andreae's published NO; and the two oxidation factors describe two different physical situations and must not be reconciled (Section 5). The bounds column exists because a published range beside a $\pm$% that could not carry it is a contradiction: such a parameter is now either drawn from those bounds (the effluent N_2O factor, bounded split-triangular in log space) or priced by named endpoint scenarios (Section Appendix D); a dash means the source states no range.

Appendix G. Supplementary Table: the IPCC sector-4 completeness matrix

Every IPCC waste category, with its notation key (E estimated, NE not estimated, IE included elsewhere) and the justification the engine carries for every non-E entry. Generated from the same completeness table the national inventory publishes, which is itself synced bidirectionally to the gap register by test – an NE row without a register entry, or a register entry without an NE row, fails the build.

Component	Status	Basis
4A First Order Decay	Tier 1, country-specific activity	IPCC Vol 5 Ch 3 FOD with default k (0.065), DOC_f, F and the uncategorised-site MCF (0.6) over a district census population panel and Pakistani generation rates (0.453 / 0.283 kg/cap/day, Pak-EPA 1996). The activity is country-specific; every decay parameter is a default, and the MCF stands in for a site-type census Pakistan has never published.
4C2 Open burning	Tier 1, country-specific activity	IPCC Eq 5.7 over the uncollected stream (0.30 urban / 0.70 rural, What a Waste) × the Box 5.1 burned fraction (0.60), with the Pak-EPA composition supplying carbon and dry matter. The burned fraction has no Pakistani measurement behind it.
4C2 air-pollutant layer	Field factors (not an IPCC tier)	Andreae 2019 garbage-burning factors on the identical burned mass. The IPCC tier ladder classifies greenhouse-gas estimation; a field-measurement factor set for co-emitted pollutants sits beside it, not on it.
4D1 CH_4	Tier 1, stratified activity	IPCC Vol 5 Ch 6 defaults (BOD 40 g/cap/day regional, B_0 0.6) over the same population panel stratified by the JMP sanitation split - the stratification is country-specific, the factors are defaults, and the sewer MCF (0.5 central, 0.1 published alongside) is the sector's deciding judgement.
4D1 N_2O	Tier 1	FAOSTAT protein through the 2019 Refinement's nitrogen chain; the effluent factor's published range spans a factor of 150, the softest number in the sector.
Provincial allocation	Country-specific allocation (not a tier)	Three distinct district-population-derived proxies (burning, disposal, wastewater load), reconciling exactly to each category's national figure. IPCC tiers classify estimation, not allocation.

Table 9: Tier status by component. Generated; the parameter values quoted are asserted against `data/waste_params.csv` at generation time.

Parameter	Value	Unit	±%	Bounds	Source (abridged)
gen_rate_urban	0.453	kg/capita/day	40	0.283–0.613	Pak-EPA / EPMC 1996 National Study on Privatization of Solid Waste Management in Eight Cities of Pakistan, Table 3 (remaining urb…
gen_rate_rural	0.283	kg/capita/day	60	–	Pak-EPA / EPMC 1996, Table 3 (rural areas)
uncollected_urban	0.3	fraction	30	–	World Bank What a Waste 2.0 (Kaza et al. 2018) Pakistan row: urban collection coverage 70%
uncollected_rural	0.7	fraction	30	–	World Bank What a Waste 2.0 (Kaza et al. 2018) Pakistan row: rural collection coverage 30%
burned_fraction	0.6	fraction	50	–	IPCC 2006 GL Vol 5 Ch 5 Box 5.1
oxidation_factor	0.71	fraction	8	–	IPCC 2019 Refinement Vol 5 Ch 5 Table 5.2 (Updated) Note 3; Yamada et al. 2010
oxidation_factor_2006	0.58	fraction	15	–	IPCC 2006 GL Vol 5 Ch 5 Table 5.2 Note 3 (US-EPA 1997)
dry_matter	0.7	fraction	20	–	Derived: Pak-EPA 1996 composition weighted by IPCC 2006 GL Vol 5 Ch 2 Table 2.4 dry-matter fractions
fossil_c_per_wet_t	0.0757	t fossil C per t wet waste	45	–	Derived: Pak-EPA 1996 composition x IPCC 2006 GL Vol 5 Ch 2 Table 2.4 (dm
total_c_per_wet_t	0.2195	t C per t wet waste	35	–	Derived: same composition and table with FCF set to 1
swds_k	0.065	1/yr	25	0.05–0.08	IPCC 2006 GL Vol 5 Ch 3 Table 3.3, bulk waste, tropical dry (MAT > 20 C, MAP < 1000 mm): default 0.065, range 0.05-0.08
swds_delay_months	6	months	–	–	IPCC 2019 Refinement Vol 5 Ch 3 (reaction delay)
swds_docf	0.5	fraction	20	–	IPCC 2006 GL Vol 5 Ch 3 s3.2.3: 'The recommended default value for DOCf is 0.5'
swds_mcf	0.6	fraction	35	–	IPCC 2006 GL Vol 5 Ch 3 Table 3.1, 'Uncategorised SWDS'
swds_f	0.5	fraction	10	–	IPCC 2006 GL Vol 5 Ch 3 s3.2.3: 'Most waste in SWDS generates a gas with approximately 50 percent CH4 ... use of the IPCC default…
swds_oxidation	0	fraction	0	–	IPCC 2006 GL Vol 5 Ch 3 Table 3.2: OX = 0 for managed, unmanaged and uncategorised SWDS
swds_recovery_t	0	tonnes CH4/yr	0	–	No landfill-gas recovery is operating in Pakistan at national scale
swds_backprojection	40	percent	–	–	Judgement: population before 2000 is back-projected from the panel's own first-decade growth rate
ww_bod_g_per_cap_day	40	g BOD/person/day	30	35–45	IPCC 2006 GL Vol 5 Ch 6 Table 6.4, 'Asia, Middle East, Latin America' (range 35-45)
ww_bod_india	34	g BOD/person/day	30	27–41	IPCC 2006 GL Vol 5 Ch 6 Table 6.4, India (range 27-41)
ww_bo	0.6	kg CH4/kg BOD	30	–	IPCC 2006 GL Vol 5 Ch 6 Table 6.2, maximum CH4 producing capacity for domestic wastewater
ww_mcf_sewer	0.5	fraction	60	0.4–0.8	IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Stagnant sewer (open and warm)' (range 0.4-0.8)
ww_mcf_sewer_river	0.1	fraction	0	0–0.2	IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Sea, river and lake discharge' (range 0-0.2)
ww_mcf_septic	0.5	fraction	0	–	IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Septic system: half of BOD settles in anaerobic tank' (range given as 0.5, no spread)
ww_mcf_latrine	0.1	fraction	50	0.05–0.15	IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Latrine: dry climate, ground water table lower than latrine, small family (3-5 persons)' (ran…
ww_mcf_open	0	fraction	0	–	IPCC 2006 GL Vol 5 Ch 6 Table 6.3 has no row for open defecation
ww_industrial_factor	1.25	fraction	20	–	IPCC 2006 GL Vol 5 Ch 6 Eq 6.3, correction factor I for additional industrial BOD discharged into sewers (default 1.25 for collec…
ww_f_npr	0.16	kg N/kg protein	10	0.15–0.17	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.10 and updated Table 6.11 (PDF p. 43): fra…
ww_f_non_con	1.1	fraction	10	1–1.4	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), updated Table 6.11 (PDF p. 43): factor for non-…
ww_f_ind_com	1.25	fraction	20	1–1.5	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.10 and updated Table 6.11 (PDF pp. 42-43):…
ww_ef_effluent	0.005	kg N2O-N/kg N	bounded split-tri. (log)	0.0005–0.075	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): freshwater, estuari…
ww_mcf_treated	0.3	fraction	0	0.2–0.4	IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Centralised, aerobic treatment plant' - not well managed / overloaded (range 0.2-0.4; well-ma…
ww_treated_share	0.01	fraction	0	0.005–0.02	Declared scenario weight; no measured national treatment share is compiled here
ww_n_sludge_kg	0	kg N/yr	–	–	IPCC 2006 GL Vol 5 Ch 6 Eq 6.8, N_SLUDGE: nitrogen removed with sludge, default zero
ww_n2o_method_2019	1	switch	–	–	This inventory's choice of method for domestic wastewater nitrous oxide
ww_f_pc	0.96	fraction	5	–	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10A (PDF p. 41): protein consumed a…
ww_n_hh	1.1	fraction	bounded split-tri. (log)	1.07–1.17	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.10 (PDF p. 40): additional nitrogen from h…
ww_f_ind_com_decentralised	1	fraction	0	–	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), text under Table 6.10B (PDF p. 42): the industr…
ww_nrem_secondary	0.4	fraction	bounded split-tri. (log)	0.35–0.55	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): nitrogen removal f…
ww_nrem_septic	0.15	fraction	bounded split-tri. (log)	0.1–0.25	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): septic tank, defau…
ww_nrem_septic_field	0.68	fraction	0	0.62–0.73	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): septic tank + land…
ww_nrem_latrine	0.12	fraction	bounded split-tri. (log)	0.07–0.21	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): latrine, default 0…
ww_ef_plant_aerobic	0.016	kg N2O-N/kg N	bounded split-tri. (log)	0.00016–0.045	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): centralised, aerobi…
ww_ef_septic	0	kg N2O-N/kg N	0	0–0.001	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): septic tank, 0 ('N2…
ww_ef_septic_field	0.0045	kg N2O-N/kg N	0	0–0.005	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): septic tank + land…
ww_ef_effluent_nutrient	0.019	kg N2O-N/kg N	0	0.0041–0.091	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39) and Section 6.3.1.2:…
ww_decentralised_to_water	1	fraction	0	0–1	IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.8 (PDF p. 42) against updated Figure 6.1 (…
ww_ef_effluent_2006_high	0.25	kg N2O-N/kg N	0	–	IPCC 2006 GL Vol 5 Ch 6 Table 6.11, emission factor for N2O from domestic wastewater effluent, upper end of the range 0.0005-0.25
waste_ch4	6.5	g CH4/kg wet waste	60	–	IPCC 2006 GL Vol 5 Ch 5 section 5.4.2 (6500 g/t wet waste; EIIP / US-EPA 2001 Vol III Ch 16 Open Burning)
waste_n2o	0.15	g N2O/kg dry waste	100	–	IPCC 2006 GL Vol 5 Ch 5 Table 5.6
waste_gef_pm25	9.7	g/kg dry matter	45	–	Andreae 2019 ACP Table 1 garbage burning (N=3)
waste_gef_co	66	g/kg dry matter	50	–	Andreae 2019 ACP Table 1 garbage burning (N=3)
waste_gef_nox	2.3	g/kg dry matter	90	–	Andreae 2019 ACP Table 1 garbage burning; published as NO: 1.5 (N=2)
waste_gef_nmvoc	23	g/kg dry matter	70	–	Andreae 2019 ACP Table 1 garbage burning total NMOGs including unidentified
waste_gef_bc	1.4	g/kg dry matter	120	–	Andreae 2019 ACP Table 1 garbage burning BC/EC (N=3)
waste_gef_so2	0.5	g/kg dry matter	80	–	Andreae 2019 ACP Table 1 garbage burning (N=1)
waste_gef_nh3	0.9	g/kg dry matter	80	–	Andreae 2019 ACP Table 1 garbage burning (N=2)
waste_gef_hcl	2.8	g/kg dry matter	65	–	Andreae 2019 ACP Table 1 garbage burning (N=3)

Table 10: Full parameter and factor provenance, generated from `data/waste_params.csv` and the `waste_burning` rows of `data/emission_factors.csv`. Each repository row carries the unabridged source and a note; a provenance test asserts that no row lacks a citation and that informally sourced values are declared as such.

Code	Category	Key	Gases	Justification (abridged)
4.A	Solid waste disposal - municipal	E	CH4	IPCC Vol 5 Ch 3 First Order Decay over a 1960-2024 deposit series, driven by the COLLECTED complement of the open-burning stream, so the two categories partition one generation figure and cannot double-count a tonne. Degradable organic carbon is the biogenic…
4.A	Solid waste disposal - industrial waste	NE	CH4	The estimated category is municipal solid waste only. Industrial solid waste reaching disposal sites has no national tonnage here; the provincial EPAs' industrial waste returns are the closing source. See waste-industrial.
4.A	Solid waste disposal - construction waste	NE	CH4	Construction and demolition waste is largely inert (near-zero degradable carbon), so its CH4 potential is small by composition - but its tonnage is unquantified and it is a distinct absence from industrial waste, with a different closing source: the Urban Uni…
4.A	Solid waste disposal - CO2 from landfill gas	IE	CO2	Biogenic, and unlike open burning IPCC does not request it as a memo item: that carbon is tracked as a stock change in AFOLU.
4.B	Biological treatment of solid waste (composting, anaerobic…	NE	CH4/N2O	Municipal composting operates at city scale - Lahore's composting facility is the largest, with smaller plants elsewhere - but no national throughput series has been compiled here. Believed small against 4.A, and the omission biases 4.A UP rather than the tot…
4.C.1	Waste incineration	NE	CO2/CH4/N2O	No municipal-scale incineration operates in Pakistan, but hospital and clinical-waste incinerators do; their throughput has not been obtained here. Bounded far below 4.C.2, whose 3.83 Mt of open burning dwarfs any clinical stream. See waste-incineration.
4.C.2	Open burning of waste	E	CO2/CH4/N2O	IPCC Eq 5.7 over the UNCOLLECTED stream x a 0.60 burned fraction. Only fossil CO2 counts; biogenic CO2 is a memo item.
4.D.1	Domestic wastewater	E	CH4/N2O	IPCC Vol 5 Ch 6 on the district population panel, stratified by the WHO/UNICEF JMP sanitation split - sewer 62.6% urban against 8.8% rural, which is why the stratification matters. THE JUDGEMENT THAT DECIDES THIS CATEGORY is the sewer MCF: IPCC gives stagna…
4.D	Industrial wastewater	NE	CH4	Needs industrial output by sector and wastewater organic loading, neither of which is available; blocked with 1.A.2 on the same HDIP/industry data.

Table 11: The waste-sector completeness matrix, generated from the engine (`outputs/completeness.csv`, sector-4 rows). Full justifications are in the generated national table; each NE row names its gap-register entry.